

Learned Body Models and Withdrawn Teachers in Muscle-Driven Motor Learning

Anonymous Author(s)

Abstract—A motor learner can be helped by a model of its own body, learned from self-generated movement, or by a more skilled teacher. Two questions about such help are rarely tested with controls: whether a body model helps because of what it has learned or merely because it adds imagined experience, and whether a learner whose teacher is withdrawn is left dependent on it. This paper answers both with controls that change one thing at a time. One soft actor-critic learner is trained on six muscle-driven bodies, an elbow and a finger from MyoSuite and four planar arms of one to four joints. A frozen body model learned from reward-free babbling is compared with a random model in identical imagination machinery; a frozen teacher is withdrawn, and compared with replacing it by a frozen copy of the student, which keeps the imitation objective and removes only the teacher. Every claim holds under two endpoints. What the body model has learned matters more the more joints the body has: the gap between the trained and the random model grows from 5 mm on one joint to 357 mm on four. On the finger, a learner whose competent teacher is withdrawn keeps at least 60% of the teacher's benefit. The sharp drop at withdrawal is lost teacher knowledge when the teacher is withdrawn early, and the abrupt change of objective when it is withdrawn later, after the student has absorbed the teacher.

Index Terms—Developmental robotics, internal models, guidance, musculoskeletal control, reinforcement learning.

I. INTRODUCTION

MUSCLE-DRIVEN bodies are hard to control: each joint is pulled by several muscles, and many activation patterns produce the same movement, so the number of degrees of freedom to coordinate grows quickly with the number of joints [1]. Deep reinforcement learning on such bodies is slow and is mostly model-free [2]–[4]. Two kinds of help are available to a learner before task learning begins. One is an internal forward model of the body, predicting the consequences of motor commands [5], [6] and learned from self-generated movement [7]; an artificial learner can use it to imagine experience. The other is a more skilled other: a teacher policy whose actions the learner imitates.

Both kinds of help raise a question that existing studies leave open. For the body model, the question is *why* it helps. Imagined rollouts add data whatever the model contains, and an inaccurate model can make model-based learning worse than model-free [8], [9]; whether a model learned from babbling helps through what it has learned, and how that depends on the body, has not been isolated. For the teacher, the question is what happens when it is taken away. Motor-learning research holds that guidance can leave a learner dependent on it [10], [11], and teacher–student pipelines in robotics cut a distillation term to make the student independent [12], [13]. Cutting the term removes the teacher's knowledge and changes the

student's objective at the same moment, so a collapse at weaning [13] cannot be attributed to either.

Both questions can be answered with a control that changes one thing at a time. A random model in the identical imagination machinery isolates what the body model has learned. A frozen copy of the student, swapped in for the teacher at withdrawal, keeps the imitation objective and removes only the teacher; comparing it with a plain withdrawal separates lost knowledge from the change of objective.

The study applies these controls to one soft actor-critic learner on six muscle-driven bodies (Fig. 1). The contributions are:

- evidence that what a babble-trained body model has learned matters more the more joints the body has: the gap between the trained and a random model grows by about half the model-free error per added joint, while a random model in the same machinery becomes increasingly harmful (Sec. IV-A);
- evidence that a competent teacher can be withdrawn without a lasting deficit: on the finger the withdrawn learner keeps at least 60% of the teacher's benefit, more the longer it was taught (Sec. IV-B);
- a decomposition of the drop at withdrawal into lost teacher knowledge and the change of objective, showing that the first dominates when the teacher is withdrawn early and the second when it is withdrawn later (Sec. IV-C).

Every claim holds under two endpoints and survives a correction for multiple comparisons. Code and data will be released on acceptance.

II. RELATED WORK

Body models. Developmental robotics [14]–[16] treats motor babbling as a way to learn forward models of the self [7], and intrinsic motivation as what drives it [17]; self-modelling [18] and goal babbling [19] learn body models by directed exploration. In model-based reinforcement learning, World Models [20] trains a model once from random rollouts; the Dreamer series [21]–[23] and TD-MPC2 [24] keep the model learning online; MBPO [8] branches short rollouts from real states. Plan2Explore [25] and APV [26] pretrain a model without task reward; offline methods [27]–[29] fit a model to logged, rewarded data. Muscle world models are co-learned with the skill [30], [31], and frozen models learned from reward-free actuation drive soft and muscle-driven bodies by planning or model-only learning [32]–[34]. That model error can make model-based learning worse than model-free,

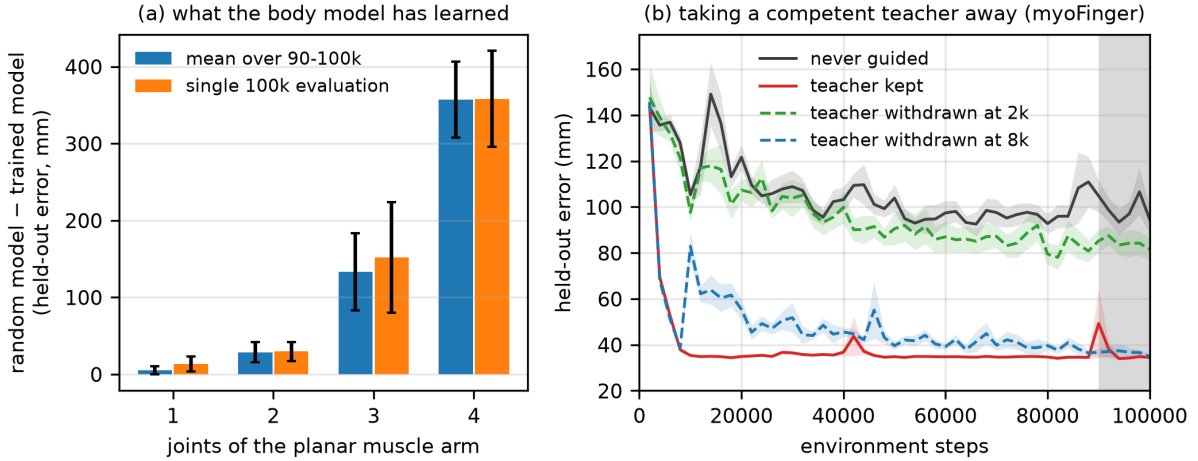


Fig. 1. The two findings. (a) What the body model has learned matters more the more joints the body has: the random model’s error minus the trained model’s error, on planar muscle arms of one to four joints, under both endpoints (bars: mean; whiskers: 95% interval; $n = 12$ on one and two joints, 24 on three and four). (b) A competent teacher can be taken away: on myoFinger, learners whose teacher is withdrawn (dashed) drop sharply, recover, and end far below a learner never guided (grey), the more so the longer they were guided. Mean across 12 seeds, band ± 1 SE; the grey band is the scored window.

more so in higher dimension, is known [8], [9]. These studies evaluate a model against model-free learning; how much of its benefit is due to what it has learned, measured against a model with no knowledge in the same machinery, and how this scales with the body, is not reported.

Guidance and its withdrawal. The guidance hypothesis [10], [11], [35] holds that frequent augmented feedback guides performance during acquisition but can degrade learning; high-frequency physical guidance gives the poorest retention in one test [36], and reduced knowledge of results improves delayed retention in another [37]. The evidence is mixed: one meta-analysis finds the degrading effect limited to laboratory tasks, parameter learning and extensive practice [38], a later one finds no reliable benefit of reduced feedback frequency [39], and children given reduced feedback learn worse [40]. A recurring lesson is that performance while guided, or just after guidance ends, is not the same as what has been learned [11], [41]. Human studies corrupt or remove the information in a feedback channel while keeping the channel [42]–[44]. In teacher–student learning, a teacher that sees privileged state is distilled into a student on the student’s own rollouts [45]–[48], and the distillation term is annealed or cut to make the student independent [12], [13]. Supervised distillation uses the student itself or a random network as the teacher to separate what the term does from what the teacher knows [49], [50]; a term toward the learner’s own earlier policy stabilises fine-tuning [51]–[53]. What is missing is a comparison of the withdrawn student with one never taught, with the teacher’s knowledge held apart from the change of objective.

III. METHOD

A. Bodies, task and learner

myoElbowPose1D6M is a one-joint elbow driven by six muscles; error is a joint-angle error in milliradians (mrad). myoFingerReachRandom-v0 is a four-joint index finger driven by five muscles; error is a Euclidean fingertip error

in millimetres (mm). Both are used with MyoSuite dynamics [54], a reach reward, a fixed start pose and eight training targets. Four planar arms of one, two, three and four revolute joints, driven by two, four, eight and ten MuJoCo muscles (antagonist pairs, biarticular from three joints on), reach a planar fingertip target; error is in mm. Joint count, muscle count and biarticular muscles therefore change together along the arm series. Episodes are 100 steps; evaluation is greedy on twenty held-out targets disjoint from the training targets. One soft actor-critic (SAC) [55], [56] learner is used in every condition: two hidden layers of 128 units, update-to-data ratio 2, batch 128, learning rate 3×10^{-4} , automatic entropy temperature. Conditions differ only in what is added. The reward, given on every step in every condition, is the only feedback a learner without help receives.

B. The body model and its control

The body model is a forward model $f_{\theta}(s, a) \mapsto s'$, a two-layer 256-unit network trained once on reward-free Ornstein–Uhlenbeck muscle activations (10^5 transitions on the elbow, 2×10^5 on the other bodies), then frozen. It is used for Dyna [57] rollouts of three steps branched from 200 real states every 100 environment steps; the reward at an imagined state is the environment’s analytic reach reward, computed with the simulator’s forward kinematics. Each update draws 64 real and 64 imagined transitions. Conditions: `prior` uses the trained model; `randprior` uses a randomly initialised model in identical machinery, so the two differ only in what the model has learned; `blank` is SAC alone with 128 real transitions per update; `blank64` uses 64 real transitions, each repeated once, the real-data content of a Dyna update without a model.

C. The teacher and its withdrawal

The teacher is a frozen actor. The student's actor loss is

$$\mathcal{L}_\pi = \mathcal{L}_{\text{SAC}} + \frac{c}{|\mathcal{A}|} \left\| \tanh \mu_{\text{student}}(o) - \tanh \mu_{\text{teacher}}(o) \right\|^2, \quad (1)$$

on replayed observations o , where $|\mathcal{A}|$ is the number of muscles; $c = 1$ while the teacher is attached. The imitation term acts on the student's actions on every update, the analogue of concurrent physical guidance rather than of knowledge of results. On the elbow the teacher (T1) is the terminal actor of a model-free run of 30k rewarded steps on a seed outside the study. On the finger, the corresponding actor barely moves (169.1 mm held-out against 172.3 for a zero-activation policy), so the finger teacher (T2) was chosen to be competent: the terminal actor of the best of twelve 100k body-model runs (35.6 mm; the twelve range from 35.6 to 85.3). One student shares its initialisation seed with that run; excluding it changes no conclusion.

At step $t_w \in \{2000, 3000, \dots, 8000\}$ the teacher is withdrawn. `none` is never guided (identical to `blank`), `constant` keeps the teacher, `abrupt` sets $c = 0$ at t_w , and `selfanchor` replaces the teacher at t_w by a frozen copy of the student's own actor, with c unchanged (the swap fires two environment steps before t_w , where the weight is last written). The withdrawn learner is compared with `none` to ask whether it is left worse off than a learner never guided. The drop at withdrawal is split by the self-anchor: `selfanchor` – `constant` is losing the teacher with the objective kept (the *teacher* component), `abrupt` – `selfanchor` is losing the imitation term (the *loss-term* component). The snapshot holds what the student has absorbed by t_w , so the teacher component measures only what the teacher still has to add.

D. Endpoints and statistics

Each run lasts 100,000 steps and is evaluated every 2,000 steps (every 1,000 in the withdrawal conditions). Every claim is reported under two endpoints: the mean of the six evaluations at 90k–100k (the *late mean*) and the single evaluation at 100k. The late mean was adopted after an internal review because single evaluations of one run move by several mrad or mm between neighbouring checkpoints; to rule out that a conclusion depends on this choice, only results that hold under both endpoints are claimed. The *drop* at withdrawal is the error at $t_w + 1000$ minus the error at t_w . The seed is the unit of analysis: twelve matched seeds per condition, twenty-four on the three- and four-joint arms, paired contrasts, paired- t 95% intervals; bold or * marks an interval excluding zero. A non-detected contrast is not evidence of no difference. Retention of a teacher's benefit is tested directly: the withdrawn learner keeps at least a fraction r of the teacher's benefit if the one-sided 95% upper bound of $(\text{abrupt} - \text{none}) - r(\text{constant} - \text{none})$, pooled over t_w within each seed, is below zero. The twelve primary tests (the trained-versus-random gaps on the two- to four-joint arms, the joint-count trend, and the finger withdrawal contrasts, each under both endpoints) all remain significant after a Holm correction (largest corrected $p = 0.043$).

E. Implementation details

Actor and critics are SiLU networks, the twin critics with layer normalisation; $\gamma = 0.99$, Polyak $\tau = 0.005$, Adam, target entropy $-0.5|\mathcal{A}|$, gradient norm clipped at 10, real replay 200,000, imagined replay 400,000, 1,000 warm-up steps. Episode time-outs are stored as terminal transitions; imagined transitions are never terminal. Babble data use $\theta = 0.15$, $\sigma = 0.4$, activations clipped to $[-1, 1]$ and the pose re-randomised every 100 steps. A matched seed sets the network initialisation, replay sampling, the training environment and the order of target draws; runs are deterministic on CPU. The 1,056 runs behind the results take about 1,100 single-thread CPU-hours.

IV. RESULTS

A. What the body model has learned matters more the more joints there are

Table I compares the trained body model with a random model in identical machinery. The two differ only in what the model has learned, and the gap grows with the number of joints (Fig. 1a): on the planar arms it is $-4.92[-10.28, +0.43]$ mm on one joint (detected only at the single 100k evaluation), $-28.42[-41.59, -15.25]$ on two, $-132.89[-183.14, -82.64]$ on three and $-357.20[-406.71, -307.69]$ on four. As a fraction of each arm's model-free error the gap is 0.05, 0.81, 0.75 and 1.78, and its per-seed slope across joint count is $-0.54[-0.69, -0.39]$ of the model-free error per joint (single evaluation: $-0.51[-0.70, -0.32]$). The trained model also beats the random one on both MyoSuite bodies, by $-25.78[-34.71, -16.84]$ mrad on the elbow and $-50.11[-64.54, -35.68]$ mm on the finger. The gap is still widening at 100k on the one-, three- and four-joint arms (by $-60.47[-101.83, -19.11]$ mm on three joints over the last 20k steps), so the reported gaps are not inflated by incomplete training.

Most of the growth comes from the random model. Imagined experience from a model that knows nothing about the body is nearly harmless on one joint ($-4.24[-6.93, -1.54]$ mm against `blank`, late mean only) and increasingly harmful after: $+22.02[+9.81, +34.23]$ mm on two joints, $+97.37[+41.17, +153.56]$ on three and $+262.20[+198.84, +325.57]$ on four (Fig. 2). With more degrees of freedom there are more ways for imagined transitions to be wrong, and the learner depends more on the model being right. Against model-free learning the trained model is ahead on the finger and on the one-, two- and four-joint arms under both baselines and both endpoints; on the elbow and the three-joint arm its advantage depends on the endpoint or the baseline (Table I). Part of its advantage over `blank` on the harder arms is the smaller real batch of a Dyna update, which helps there by itself; the comparison with the random model, which shares that batch, is free of this effect.

B. A competent teacher can be taken away

On the finger, with a competent teacher, the withdrawn learner ends far ahead of a learner never guided

TABLE I

WHAT THE BODY MODEL HAS LEARNED, ON SIX BODIES: LATE-MEAN ERROR OF THE TRAINED BODY MODEL MINUS A RANDOM MODEL IN IDENTICAL MACHINERY, AND MINUS THE TWO MODEL-FREE BASELINES, WITH 95% INTERVALS. BOLD: INTERVAL EXCLUDING ZERO. †: ALSO EXCLUDING ZERO, WITH THE SAME SIGN, AT THE SINGLE 100K EVALUATION. NEGATIVE FAVOURS THE TRAINED MODEL.

Body	n	trained – random model	trained – blank	trained – blank64
myoElbow (mrad)	12	-25.8 [†] [-34.7, -16.8]	-3.9 [-7.4, -0.4]	-7.8 [-11.0, -4.6]
myoFinger (mm)	12	-50.1 [†] [-64.5, -35.7]	-46.2 [†] [-58.7, -33.7]	-25.9 [†] [-33.8, -18.0]
1-joint arm (mm)	12	-4.9 [-10.3, +0.4]	-9.2 [†] [-15.0, -3.3]	-11.7 [†] [-21.8, -1.7]
2-joint arm (mm)	12	-28.4 [†] [-41.6, -15.3]	-6.4 [†] [-11.1, -1.7]	-9.8 [†] [-16.7, -2.9]
3-joint arm (mm)	24	-132.9 [†] [-183.1, -82.6]	-35.5 [-62.3, -8.7]	-7.8 [-30.1, +14.4]
4-joint arm (mm)	24	-357.2 [†] [-406.7, -307.7]	-95.0 [†] [-134.0, -56.0]	-35.3 [†] [-59.5, -11.1]

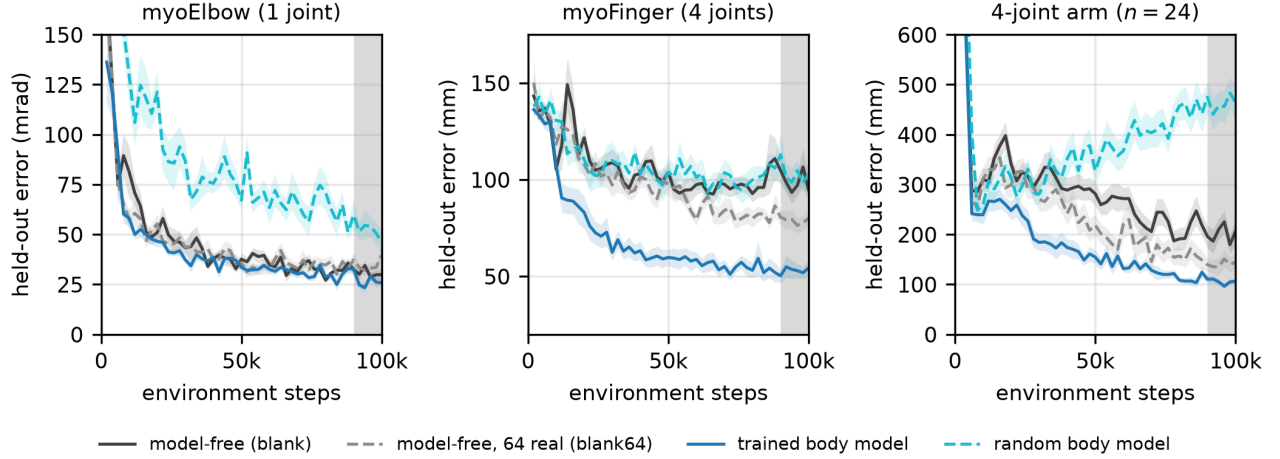


Fig. 2. Learning curves with the trained body model (blue), a random body model in the same machinery (cyan, dashed) and the two model-free baselines (grey). On the four-joint arm the random model ends far above model-free learning while the trained model ends far below it. Held-out error, mean, band ± 1 SE; grey band: the scored window.

TABLE II

TAKING THE TEACHER AWAY, POOLED OVER THE SEVEN WITHDRAWAL TIMES ($t_w = 2000, \dots, 8000$), $n = 12$, UNDER BOTH ENDPOINTS. WORTH: CONSTANT-NONE, WHAT KEEPING THE TEACHER GIVES OVER NEVER BEING GUIDED. WITHDRAWN-NEVER: ABRUPT-NONE. KEPT: THE LARGEST FRACTION OF THE TEACHER'S WORTH THAT THE WITHDRAWN LEARNER PROBABLY KEEPS (ONE-SIDED 95%). BOLD: INTERVAL EXCLUDING ZERO.

	late mean (90–100k)			single 100k evaluation		
	worth	withdrawn—never	kept	worth	withdrawn—never	kept
myoElbow (mrad), T1	-8.6 [-12.3, -5.0]	+1.8 [-1.4, +4.9]	—	-8.1 [-15.8, -0.3]	+3.4 [-5.6, +12.4]	—
myoFinger (mm), T2	-61.7 [-75.0, -48.4]	-43.3 [-54.7, -31.8]	$\geq 62\%$	-60.2 [-73.5, -46.8]	-40.6 [-53.0, -28.2]	$\geq 60\%$

(Table II, Fig. 1b): $-43.25 [-54.72, -31.79]$ mm pooled over the seven withdrawal times (single evaluation: $-40.63 [-53.05, -28.22]$), against a teacher worth $-61.71 [-74.98, -48.44]$. It provably keeps at least 62% of the teacher's benefit (60% at the single evaluation). It keeps more the longer it was taught: the difference from the never-guided learner grows from -14.82 mm at $t_w = 2000$ to -62.60 at $t_w = 8000$, and from $t_w = 7000$ on the withdrawn learner is not detectably behind one that kept its teacher, under either endpoint.

On the elbow the teacher is worth less, $-8.65 [-12.25, -5.05]$ mrad, and its benefit goes with it: the withdrawn learner ends 9 to 12 mrad behind one that kept its teacher at every t_w , and is not detectably different from one never guided ($+1.76 [-1.36, +4.89]$; single evaluation $+3.42 [-5.58, +12.42]$). With a benefit

this small, 100k steps cannot exclude a small deficit at the single evaluation. Running the elbow conditions to 200k on the same seeds (their first 100k reproduce the runs above at every checkpoint) narrows it: over 190–200k the withdrawn learner is $+0.04 [-2.11, +2.19]$ mrad from one never guided (single 200k evaluation: $-0.51 [-5.20, +4.18]$), and under both endpoints any deficit is smaller than the teacher's worth (one-sided 95%).

C. The drop at withdrawal: lost knowledge early, the change of objective later

Withdrawal is followed by a sharp rise in error (Fig. 1b), of $+42.98 [+8.71, +77.26]$ mm to $+72.54 [+58.26, +86.82]$ on the finger and $+32.05 [+12.70, +51.41]$ mrad to $+80.56 [+23.00, +138.12]$ on the elbow over the next 1,000 steps relative to keeping the teacher (Table III, Fig. 3).

TABLE III

THE RISE IN ERROR OVER THE 1,000 STEPS AFTER WITHDRAWAL AT t_w , AGAINST A LEARNER THAT KEEPS ITS TEACHER (TOTAL), SPLIT BY THE SELF-ANCHOR INTO THE TEACHER COMPONENT AND THE LOSS-TERM COMPONENT; PAIRED MEANS, $n = 12$, *: INTERVAL EXCLUDING ZERO.

		2k	3k	4k	5k	6k	7k	8k
myoElbow (mrad)	total	+59.5*	+39.7*	+63.5*	+44.1*	+80.6*	+32.1*	+45.2*
	teacher	+46.0*	+22.0*	+25.6*	+1.5	+2.7	+7.8	+5.0
	loss term	+13.5	+17.6*	+37.9*	+42.6	+77.9*	+24.3*	+40.2*
myoFinger (mm)	total	+43.0*	+45.6*	+57.5*	+60.7*	+72.5*	+62.7*	+49.2*
	teacher	+35.7*	+23.6*	+17.2*	+21.4*	+16.1*	+8.1*	+3.0*
	loss term	+7.3	+22.0	+40.2*	+39.3*	+56.4*	+54.6*	+46.2*

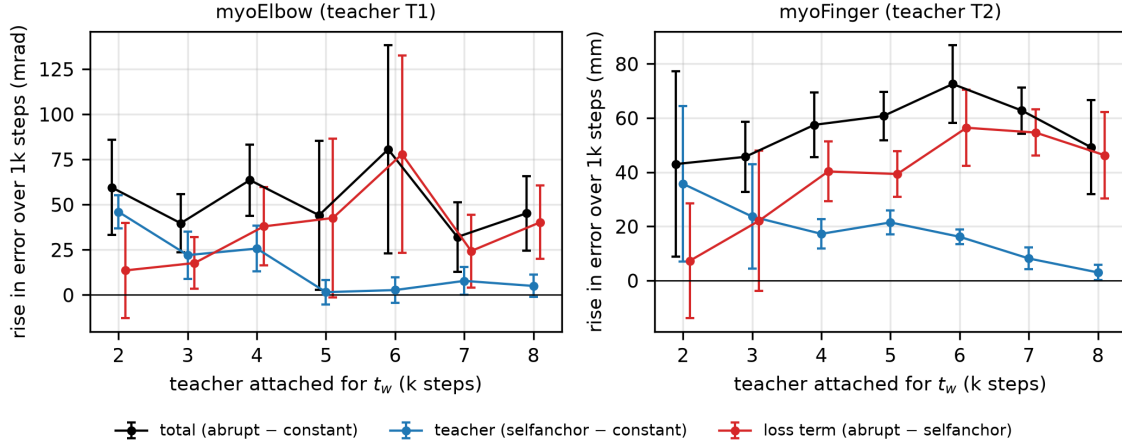


Fig. 3. The drop at withdrawal against the time the teacher was attached (Table III). Early withdrawal costs mostly the teacher’s knowledge (blue); later withdrawal costs mostly the imitation objective (red), once the student has absorbed what the teacher had to give. Paired means with 95% intervals, $n = 12$.

What it consists of depends on how long the teacher was attached. Withdrawn at $t_w = 2000$, the learner loses mostly the teacher’s knowledge: the teacher component is $+45.97 [+36.74, +55.20]$ mrad on the elbow, larger than the loss-term component by $+32.42 [+3.09, +61.75]$, and $+35.69 [+7.06, +64.32]$ mm on the finger. Withdrawn later, the drop is mostly the change of objective. On the finger the loss-term component exceeds the teacher component at every t_w from 4000, by 17.87 to 46.48 mm; on the elbow the teacher component is no longer detected from $t_w = 5000$ on, and the loss-term component exceeds it at $t_w = 6000$ and 8000. The teacher component falls as attachment lengthens because the self-anchor holds what the student has already absorbed; this fall therefore reflects absorption by construction, and the new information is the size of what remains: the drop that follows a late withdrawal is a reaction to the objective changing, and a term toward the student’s own snapshot removes most of it.

V. DISCUSSION

A body model is worth what it has learned. Imagination by itself is not what helps. With a model that knows nothing about the body, imagined experience costs little on one joint and becomes destructive as the number of joints grows; with a model learned from babbling, the same machinery helps. The difference between the two is the value of what babbling taught the model, and it grows with the degrees of freedom the learner has to coordinate [1]. For a developing learner,

this is an argument that self-generated movement pays most on the bodies that are hardest to control, and that a wrong body model on such a body is worse than none.

Taking a teacher away. In the human sense, the sharp drop after withdrawal is the retention-test analogue, and it is real: guidance does leave the learner unable to perform at the guided level immediately afterwards [36]. What it does not leave, after continued practice, is a deficit in what has been learned [11], [41]: on the finger most of a competent teacher’s benefit persists, and the withdrawn learner is never detectably behind one never guided on either body. The self-anchor shows when the drop reflects learning and when it reflects performance: early, the student has not yet absorbed the teacher and loses its knowledge; later, the student has absorbed it, and the drop is the objective changing under it. The setting differs from the human studies: the teacher’s attachment duration is varied rather than feedback frequency, and reward is given on every step in every condition. A practical consequence follows for pipelines that wean a student off distillation [12], [13]: a late withdrawal can be smoothed by anchoring the imitation term to the student’s own snapshot rather than cutting it.

VI. LIMITATIONS AND FUTURE WORK

The study is single-task reach in simulation, on deterministic, fully observed bodies, which favour a frozen forward model, and the imagined reward uses exact forward kinematics; partially observed bodies and a learned reward are the

natural next test. Along the arm series, joint count, muscle count and biarticular muscles change together, so the scaling is with the body's complexity as a whole rather than with joints alone. Each random control is one initialisation per seed. The finger's competent teacher is the best of twelve candidate runs; teachers of graded quality would map the benefit against competence. Withdrawal is a single step at 2k–8k steps; gradual fading and late withdrawal are left to future work, as are longer training on the harder bodies and a physical robot.

VII. CONCLUSION

Two kinds of help for a muscle-driven learner are tested with controls that change one thing at a time. A body model learned from babbling helps through what it has learned, and that matters more the more joints the body has: a random model in the same machinery turns from harmless to destructive as joints are added. A competent teacher can be taken away: on the finger the learner keeps most of what it was taught, and the drop at withdrawal is lost knowledge only when the teacher leaves early; later it is the change of objective.

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